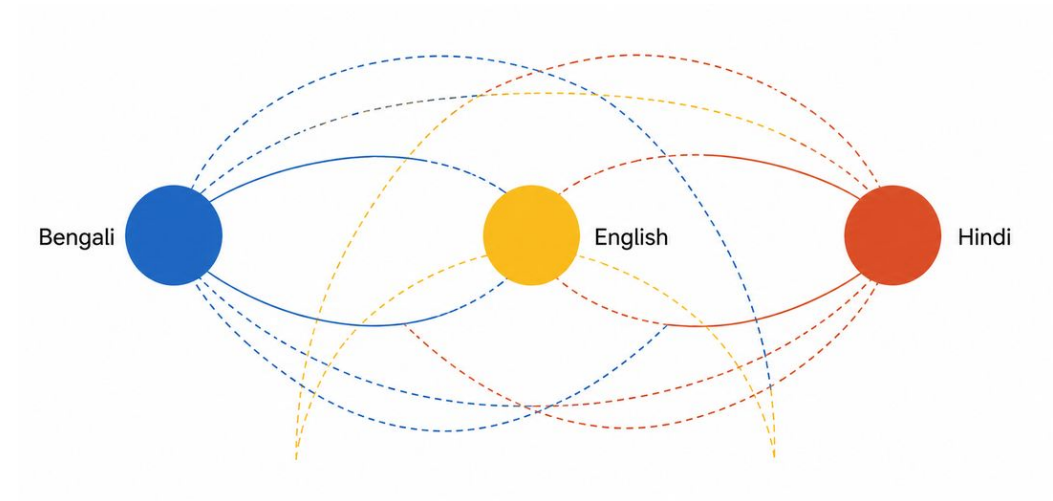


Cross Lingual Bengali Idiom Matching Using Multilingual Sentence Embeddings Without Translation - Integrating *BanglaBERT* and *XLM-RoBERTa*



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Introduction: The Challenge of Idioms

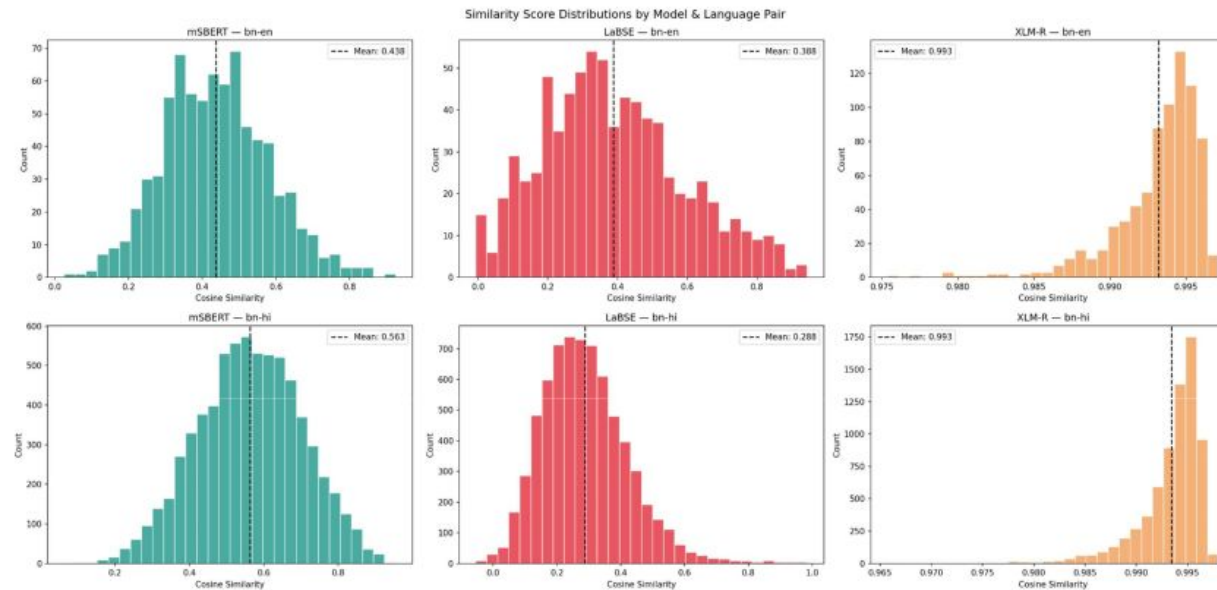


Fig 5.1: Similarity Score Distributions by Model and Language Pair

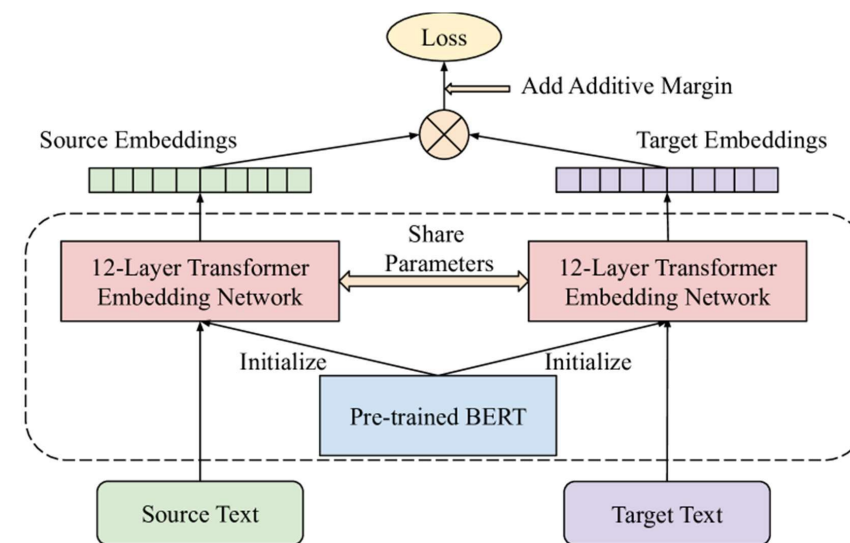
Idioms are non-compositional: meaning does not follow from constituent words. Example: Bengali "আকাশ কুসুম" literally 'sky-flower' → idiomatic: impossible dream. We ask: can pretrained embeddings align such idioms across Bengali, English, and Hindi without any translation layer or task-specific fine-tuning?



Core constraint: All matching is via cosine similarity in a shared embedding space — no MT.

Related Work — Foundations

Paper	Year	Relevance
Attention Is All You Need - Vaswani et al.	2017	Transformer architecture foundation
Neural Machine Translation - Bahdanau et al.	2014	Alignment via attention
BERT - Devlin et al.	2019	Contextualized token representations
Sentence-BERT - Reimers & Gurevych	2019	Siamese sentence encoders
LaBSE - Feng et al.	2022	Dual-encoder for multilingual sentence embeddings
XLM-R - Conneau et al.	2020	Large multilingual MLM pretraining
BanglaBERT - Bhattacharjee et al.	2022	Monolingual Bengali representation power



These works supply the architectural and methodological building blocks: transformers, sentence encoders, and multilingual pretraining. We evaluate how these precepts transfer to idiom matching without translation.

Dataset Construction — Aligning Bengali, English, Hindi

Sources compiled and harmonized:

- Bagdhara: 8,800 Bengali idioms with human-curated `similar_in_english`
- Bengali-Bangla JSON: 6 idioms with English meanings
- English JSON: 14,200+ idioms with `figurative_meaning`
- Hindi JSON: 450+ idioms (English figurative meanings)

Alignment strategies:

- High confidence bn-en: direct `similar_in_english` mappings
- Medium confidence bn-hi: English meaning keyword overlap ($\text{jaccard} > 0.15$)

 Final tally: 9,061 bn-en pairs, 4,096 bn-hi pairs → 23,544 unique embeddings after deduplication.

Pair Type	Count & Method
bn-en	9,061 - Human-curated <code>similar_in_english</code>
bn-hi	4,096 - Keyword overlap on English meanings

Example JSON snippet:

```
{"bn_idiom": "আকাশ কুসুম", "similar_in_english": "build castles in the air"}
```

The Four Models — Training Paradigms

mSBERT

Multilingual sentence embedder -
paraphrase distilled,
English-anchored;
outputs ready
sentence vectors.

LaBSE

Dual-encoder trained
on translation pairs
(109 languages);
margin softmax
ranking strength.

Notes:

- All models used off-the-shelf (no fine-tuning).
- mSBERT & LaBSE produce sentence embeddings directly.
- BanglaBERT & XLM-R require mean pooling over token embeddings to yield sentence vectors.

XLM-R

Large multilingual
MLM without
sentence objective -
requires pooling for
sentence vectors.

BanglaBERT

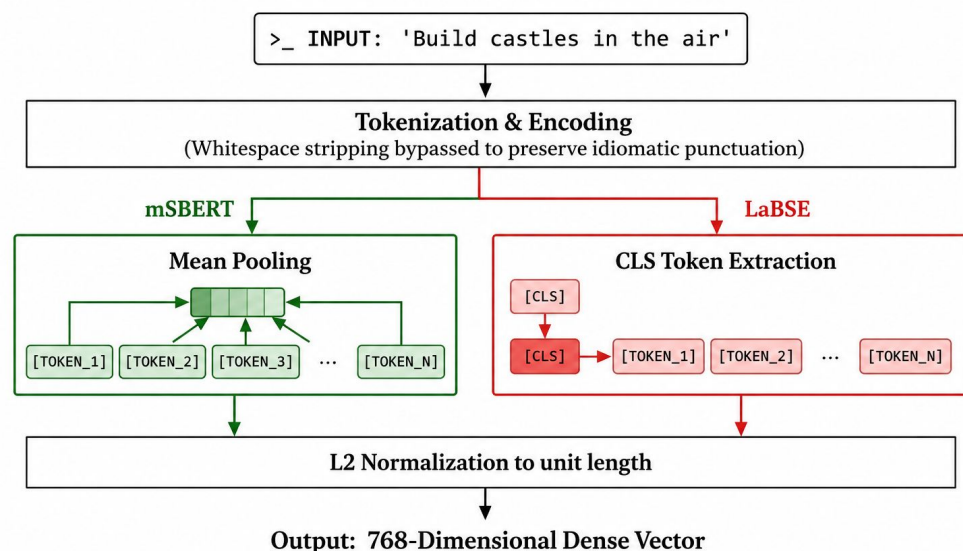
Monolingual Bengali
BERT - strong Bengali
knowledge, lacks
built-in cross-lingual
sentence mapping.



Design implication: pooling strategy quality and normalization are decisive when repurposing MLMs for sentence tasks.

Mean Pooling - Enabling BanglaBERT & XLM-R

The Embedding Generation Mechanics



BanglaBERT and XLM-R produce token-level outputs ($\text{seq_len} \times 768$). We apply mean pooling across real tokens (using attention mask) to produce a fixed 768-dim sentence vector, followed by L2 normalization before cosine.

Implementation details

Batch size = 16 (memory constrained); mask-aware averaging; final vectors normalized for cosine similarity.

Why it matters

Without mean pooling, MLMs lack a straightforward sentence representation - pooling quality impacts cross-lingual alignment.

Evaluation Tasks - Three Lenses

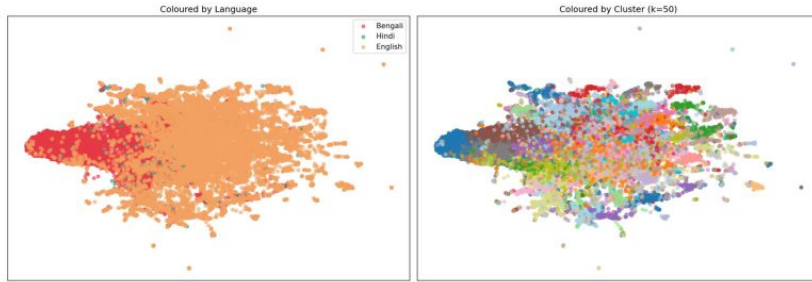


Fig 5.2: Cross-Lingual Idiom Clustering - mSBERT

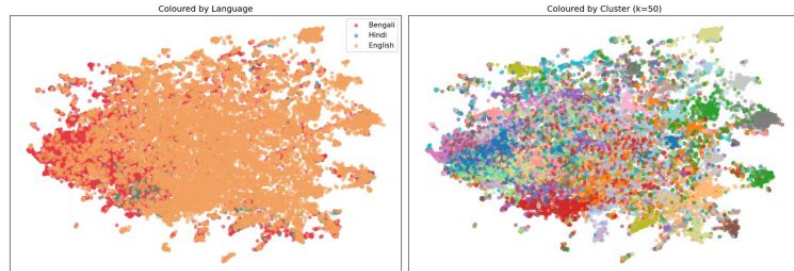


Fig 5.3: Cross-Lingual Idiom Clustering - LaBSE

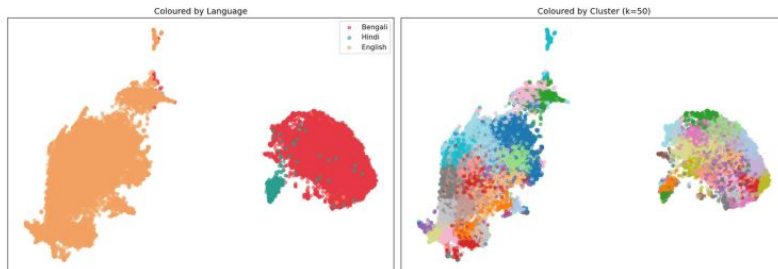


Fig 5.4: Cross-Lingual Idiom Clustering - XLM-R

- 1. **Semantic Similarity Scoring**

Cosine similarity on ground-truth pairs (bn-en, bn-hi). Report means and distributions.

- 2. **Cross-Lingual Retrieval**

Bengali query ranks English/Hindi idioms. Metrics: MRR, P@1, P@5, Hit@10.

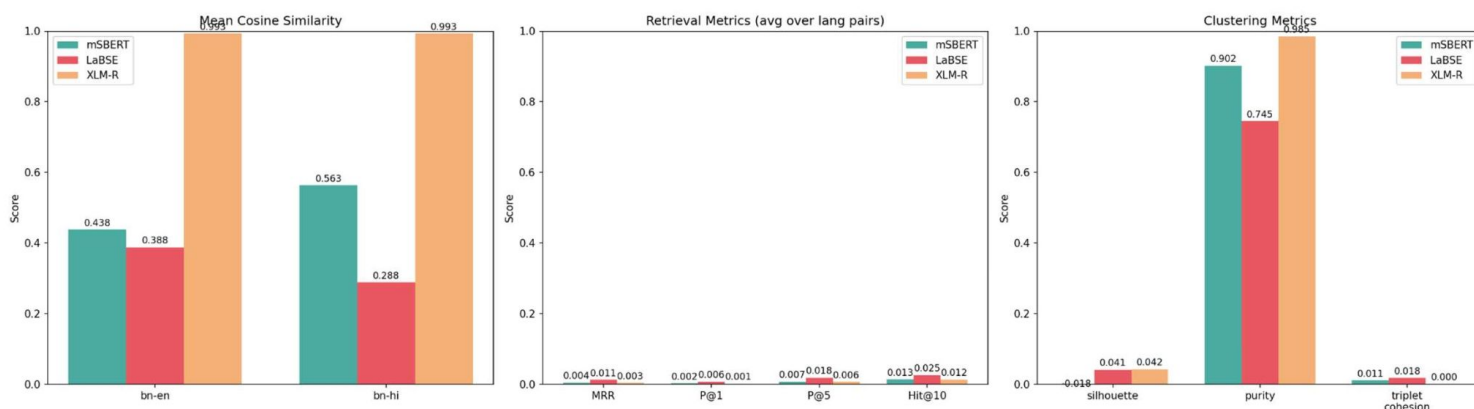
- 3. **Cross-Lingual Clustering**

K-Means (k=50) on 23k+ embeddings. Metrics: Silhouette, Purity, Triplet Cohesion (bn-en-hi in same cluster).



Triplet Cohesion is the strictest test of semantic equivalence across three languages.

Key Results — Similarity & Retrieval



Observations: mSBERT captures Indo-Aryan proximity (bn-hi > bn-en). LaBSE has lower raw similarity but excels in retrieval ranking due to its training objective. XLM-R exhibits near-1.0 similarities with extremely low variance ($\sigma \approx 0.003$) - a collapse artifact rendering scores meaningless.

Retrieval (MRR)

LaBSE 0.0115

mSBERT 0.0042

LaBSE's dual-encoder + margin softmax provides superior ranking (MRR). All models outperform a random baseline - evidence of genuine learned alignment despite differences.

⚠ XLM-R's collapse requires caution: high numeric similarity does not imply semantic correctness.

Key Results - Clustering & Triplet Cohesion

Model	Silhouette	Purity / Triplet Cohesion
mSBERT	-0.018	Purity 0.902 - Triplet 1.06%
LaBSE	+0.041	Purity 0.745 - Triplet 1.76%
XLM-R	+0.042	Purity 0.985 - Triplet 0.00%



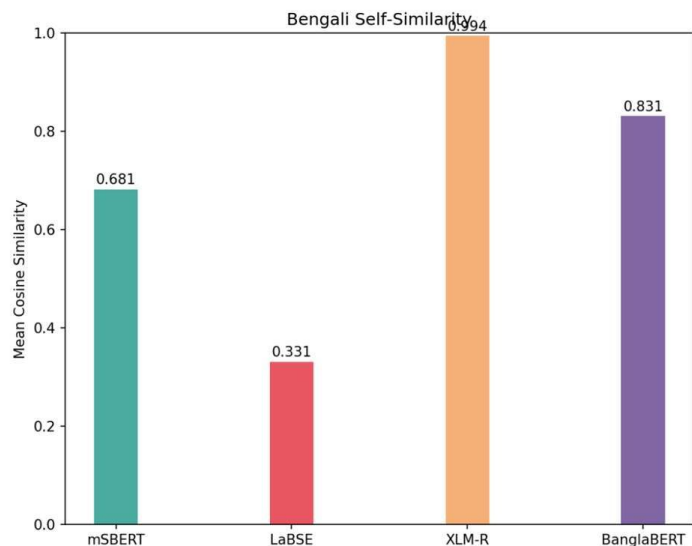
Triplet Cohesion is the most demanding metric - it requires bn, en, and hi equivalents to co-locate within the same cluster.

Interpretation:

- LaBSE: most language-agnostic - lower purity (more mixed clusters) and highest triplet cohesion (best cross-lingual grouping).
- mSBERT: clusters strongly by language (high purity), so cross-lingual triplets are rare though semantic signals exist.
- XLM-R: effectively separates languages - high purity but zero triplet cohesion, confirming collapse into language islands.

Final analytic takeaway: LaBSE's training objective gives it an edge for cross-lingual grouping and retrieval, despite lower absolute similarity scores.

Monolingual Bengali Evaluation - The Collapse Discovery



The chart above illustrates the mean intra-Bengali (bn-bn) similarity scores for each model. **XLM-R exhibits an alarmingly high mean similarity of 0.994**, which indicates a severe embedding collapse where all Bengali phrases are mapped to nearly identical vectors, making it unusable for distinguishing meaning.

Silhouette Scores (K=887)

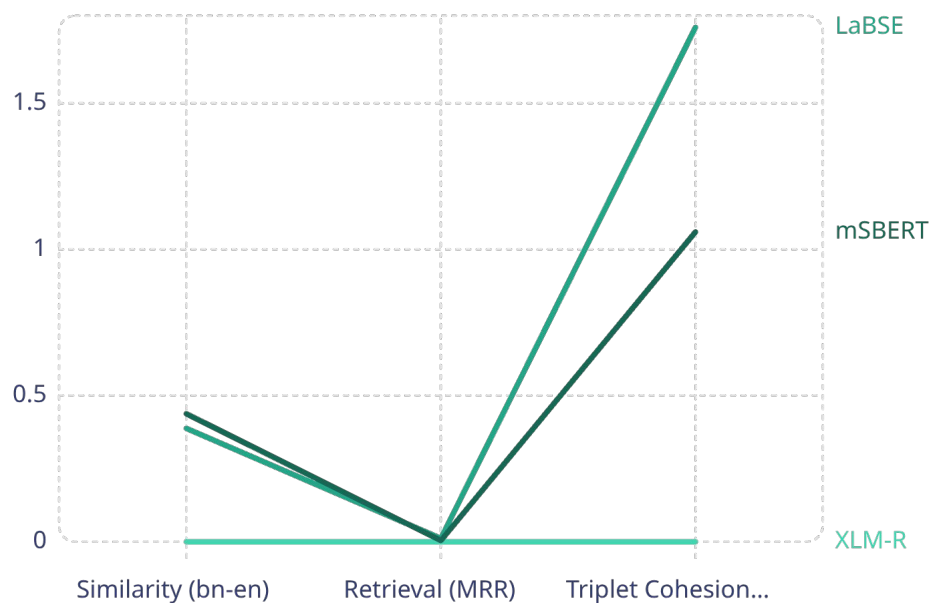
Model	Silhouette Score
LaBSE	+0.034
mSBERT	Negative
BanglaBERT	Negative
XLM-R	Negative

Only LaBSE registered a positive Silhouette score in monolingual Bengali clustering, suggesting it forms more coherent clusters. The negative scores for other models further highlight their struggle with fine-grained semantic differentiation within Bengali.



Conclusion: XLM-R suffers catastrophic embedding collapse for Bengali, rendering it unusable for fine-grained tasks. BanglaBERT is strong for monolingual Bengali but lacks inherent cross-lingual capabilities for this problem.

Final Verdict - LaBSE Wins Overall



The radar chart summarizes model performance across key cross-lingual idiom matching tasks. LaBSE demonstrates balanced strength across retrieval and clustering, while mSBERT shows higher raw similarity.

🏆 LABSE: CROSS-LINGUAL WINNER

Overall, **LaBSE** is the cross-lingual winner, showing favorable metrics in 6 categories compared to mSBERT's 0 and XLM-R's 1.

- **Use LaBSE** for retrieval (search, translation assistance) or cluster mixing.
- **Use mSBERT** if similarity confidence or linguistic proximity (bn-hi > bn-en) detection is crucial.
- **Avoid XLM-R** for any fine-grained cross-lingual idiom task due to collapse.
- **BanglaBERT** excels for monolingual Bengali tasks only.



No single model is universally best – selection is entirely task-dependent.

Drawbacks & Future Scope

Drawbacks



Dataset imbalance (Bengali 8.8k, Hindi ~200)



bn-hi ground truth noise (78% accuracy)



No fine-tuning (off-the-shelf only)



Isolated idioms (not in sentences)



Fixed K-Means (k=50)



No translation-based baseline

Future Scope



Expand Hindi idioms to 1k+ entries



Human verification of triplets → gold standard



Fine-tune with contrastive/triplet loss



Contextualised evaluation (idioms in full sentences)



Experiment with different clustering algorithms



Add MT + monolingual SBERT as control

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